

Additional file 1: Supplementary tables

No Single Model Family Dominates: Ensembles and Conventional Machine Learning Perform Comparably to Pretrained Molecular Models Across 44 Property-Prediction Benchmarks

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This file contains the eleven supplementary tables referenced in the main text. Tables S1–S7 are generated directly from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`; Table S8 is generated by `python -m qsarena.reliability_tables` from the reliability-study artifacts; Tables S9–S11 are generated by `qsarena.meta_analysis`, which the same render script runs. They carry the same values as the machine-readable CSV and Markdown files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

dataset	suite	task_ - kind	family	model	analysis_ - analysis_metric_value	cv_ - selected_ - model	cv_ - selected_ - family	cv_selected_gap_to_best	
tdc_ames	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ - roc_ - auc	0.883	Extra trees	Conventional ML	1.386
tdc_bbb_ - martins	TDC	classification	Conventional ML	Random forest	test_ - roc_ - auc	0.925	XGBoost (ADMETboost features)	Conventional ML	0.963
tdc_ - bioavailability_ - ma	TDC	classification	Conventional ML	CatBoost	test_ - roc_ - auc	0.777	XGBoost (ADMETboost features)	Conventional ML	7.129
tdc_ - carcinogens_ - lagunin	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ - roc_ - auc	0.929	XGBoost (ADMETboost features)	Conventional ML	2.237
tdc_clintox	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_ - roc_ - auc	0.974	AdaBoost	Conventional ML	8.986
tdc_cyp1a2_ - veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ - roc_ - auc	0.974	SVC	Conventional ML	1.196
tdc_cyp2c19_ - veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ - roc_ - auc	0.938	XGBoost	Conventional ML	1.338
tdc_cyp2c9_ - substrate_ - carbonmangels	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ - auprc	0.438	SVC	Conventional ML	31.368
tdc_cyp2c9_ - veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ - auprc	0.827	AdaBoost	Conventional ML	14.709
tdc_cyp2d6_ - substrate_ - carbonmangels	TDC	classification	Chemprop v2 GNN	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	test_ - auprc	0.693	ChemML MLP (PyTorch)	Deep tabular NN (ChemML MLP)	26.129
tdc_cyp2d6_ - veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ - auprc	0.763	AdaBoost	Conventional ML	19.417
tdc_cyp3a4_ - substrate_ - carbonmangels	TDC	classification	Conventional ML	XGBoost (ADMETboost features)	test_ - auprc	0.729	XGBoost (ADMETboost features)	Conventional ML	0.000
tdc_cyp3a4_ - veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_ - auprc	0.905	AdaBoost	Conventional ML	11.894

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dataset	suite	task - kind	family	model	analysis_ metric	analysis_metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_dili	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ roc_ auc	0.933	XGBoost (ADMETboost features)	Conventional ML	1.072
tdc_herg	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.857	XGBoost (ADMETboost features)	Conventional ML	7.662
tdc_herg_karim	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ roc_ auc	0.907	XGBoost	Conventional ML	0.494
tdc_hia_hou	TDC	classification	Conventional ML	XGBoost (ADMETboost features)	test_ roc_ auc	0.994	XGBoost (ADMETboost features)	Conventional ML	0.000
tdc_pampa_ ncats	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ roc_ auc	0.763	Random forest	Conventional ML	10.598
tdc_pgp_ broccatelli	TDC	classification	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ roc_ auc	0.933	Random forest	Conventional ML	3.065
tdc_skin_ reaction	TDC	classification	Conventional ML	XGBoost (ADMETboost features)	test_ roc_ auc	0.667	XGBoost (ADMETboost features)	Conventional ML	0.000
tdc_tox21	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.597	Extra trees	Conventional ML	8.142
tdc_toxcast	TDC	classification	Conventional ML	HistGradientBoosting	test_ roc_ auc	0.716	XGBoost (ADMETboost features)	Conventional ML	1.662
chemml_cep_ homo	ChemML	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.090	TabPFNRegressor	TabPFN (tabular foundation)	13.888
chemml_ organic_density	ChemML	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.005	TabPFNRegressor	TabPFN (tabular foundation)	0.000
esol_delaney	Molecule Net	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	0.622	TabPFNRegressor	TabPFN (tabular foundation)	8.211
freesolv_sampl	Molecule Net	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.933	XGBoost (ADMETboost features)	Conventional ML	121.064
lipophilicity	Molecule Net	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.507	TabPFNRegressor	TabPFN (tabular foundation)	17.351
poduam_pod_ nc_std	PODUAM	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.694	XGBoost (ADMETboost features)	Conventional ML	5.048

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dataset	suite	task - kind	family	model	analysis_ metric	analysis_metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
poduam_pod_ rd_std	PODUAM	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.556	XGBoost (ADMETboost features)	Conventional ML	1.824
polaris_adme_ fang_hppb_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.422	XGBoost (ADMETboost features)	Conventional ML	9.517
polaris_adme_ fang_perm_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.379	XGBoost (ADMETboost features)	Conventional ML	16.319
polaris_adme_ fang_rclint_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.484	XGBoost	Conventional ML	11.804
polaris_adme_ fang_rppb_1	Polaris	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.478	XGBoost (ADMETboost features)	Conventional ML	6.138
polaris_adme_ fang_solu_1	Polaris	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	0.551	XGBoost (ADMETboost features)	Conventional ML	7.782
tdc_caco2_wang	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.338	XGBoost (ADMETboost features)	Conventional ML	11.813
tdc_clearance_ hepatocyte_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	44.113	XGBoost (ADMETboost features)	Conventional ML	6.388
tdc_clearance_ microsome_az	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	33.507	XGBoost (ADMETboost features)	Conventional ML	13.113
tdc_half_life_ obach	TDC	regression	Uni-Mol (3D pretrained)	Uni-Mol V1	test_ rmse	19.128	XGBoost (ADMETboost features)	Conventional ML	77.574
tdc_ hydrationfreeenergy_ freesolv	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse OOF error))	test_ rmse	1.075	XGBoost (ADMETboost features)	Conventional ML	5.821
tdc_ld50_zhu	TDC	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_ rmse	0.806	XGBoost	Conventional ML	4.826
tdc_ lipophilicity_ astrazeneca	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.551	TabPFNRegressor	TabPFN (tabular foundation)	8.398
tdc_ppbr_az	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	10.768	TabPFNRegressor	TabPFN (tabular foundation)	21.078
tdc_solubility_ aqsoldb	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	0.955	XGBoost (ADMETboost features)	Conventional ML	4.572

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_ value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_vdss_ lombardo	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	4.670	XGBoost (ADMETboost features)	Conventional ML	69.878

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

stage_order	stage	datasets_evaluated	datasets_improved_vs_previous	improvement_fraction
1	Conventional ML only	44	0	0.000
2	+ Map Light classic fea- tures	44	9	0.205
3	+ neu- ral/ deep back- ends	44	26	0.591
4	+ CFA fusion	44	6	0.136
5	Full pipeline incl. ensem- bles	44	22	0.500

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

Feature family	Datasets selected	Available (sum)	Selected (sum)	% of selected	Enrichment vs uniform
rdkit	40	8,261	1,362	11.22	5.16
erg	38	12,552	1,146	9.44	2.86
avalon	43	44,384	3,603	29.69	2.54
maccs	38	5,962	224	1.85	1.18
maplight	41	30,070	854	7.04	0.89
fcfp6	43	42,500	998	8.22	0.73
atom_pair	41	39,651	830	6.84	0.66
rdk_path	40	45,036	896	7.38	0.62
topological_torsion	40	23,653	421	3.47	0.56
ecfp6	37	43,299	731	6.02	0.53
layered	40	43,642	674	5.55	0.48
morgan	38	40,827	398	3.28	0.31

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

model	datasets_attempted	datasets_valid
Uni-Mol V2 (310m)	8	0
Uni-Mol V2 (164m)	4	0
Uni-Mol V2 (84m)	44	17
TabPFNClassifier	22	19
ElasticNetCV	22	20
LogisticRegression	22	22
MapLight CatBoost (Strict Parity)	22	22
TabPFNRegressor	22	22
SVC	22	22
SVR	22	22
Voting Classifier (KNN, SVM)	22	22
Voting Regressor (KNN, SVM)	22	22
Tabular CNN	22	22
Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	44	38
Chemprop v2 (AttentiveFP, ensemble=3)	44	38
Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3)	44	42
Chemprop v2 (D-MPNN, ensemble=3)	44	42
Chemprop v2 (CMPNN, ensemble=3)	44	42
HistGradientBoosting	44	44
Ensemble (OOF Stacking (RidgeCV, 5-fold))	44	44
ChemML MLP (PyTorch)	44	44
CatBoost	44	44
CFA (Combinatorial Fusion)	44	44
ChemML MLP (TensorFlow)	44	44
AdaBoost	44	44
Extra trees	44	44
Ensemble (Weighted average (inverse OOF error))	44	44
Random forest	44	44
MapLight + GNN (CatBoost, Strict Parity)	44	44
Uni-Mol V1	44	44
Tabular MLP	44	44
XGBoost	44	44
XGBoost (ADMETboost features)	44	44

Table S5: Run-to-run comparison: the canonical NSF ACCESS Jetstream2 A100 benchmark against the earlier consumer-GPU (RTX 4060) run, analysed identically. “Same split” marks datasets where both runs used the identical held-out partition; the remainder were re-split to scaffold splits in the A100 run and are therefore not directly comparable. Change is relative and metric-direction aware (positive favours the A100 run).

Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_ames	classification	True	TabPFNClassifier	0.876	XGBoost	0.875	0.166
tdc_bbb_martins	classification	True	Random forest	0.925	Random forest	0.932	-0.788
tdc_bioavailability_ma	classification	True	CatBoost	0.777	CatBoost	0.777	0.000
tdc_cyp1a2_veith	classification	True	XGBoost	0.969	Uni-Mol V1	0.969	-0.049
tdc_cyp2c19_veith	classification	True	XGBoost	0.926	Uni-Mol V1	0.921	0.551
tdc_cyp2c9_substrate_carbonmangels	classification	True	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	Uni-Mol V1	0.454	-3.439
tdc_cyp2c9_veith	classification	True	XGBoost	0.796	HistGradientBoosting	0.792	0.541
tdc_cyp2d6_substrate_carbonmangels	classification	True	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	Random forest	0.665	4.143
tdc_cyp2d6_veith	classification	True	XGBoost	0.729	XGBoost	0.722	0.904
tdc_cyp3a4_substrate_carbonmangels	classification	True	XGBoost (ADMETboost features)	0.729	LogisticRegression	0.717	1.759
tdc_cyp3a4_veith	classification	True	XGBoost (ADMETboost features)	0.883	XGBoost	0.883	0.002
tdc_dili	classification	True	XGBoost (ADMETboost features)	0.923	Uni-Mol V1	0.914	0.903
tdc_herg	classification	True	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	AdaBoost	0.861	-0.411
tdc_herg_karim	classification	True	XGBoost	0.902	Uni-Mol V1	0.893	1.055
tdc_hia_hou	classification	True	XGBoost (ADMETboost features)	0.994	CatBoost	0.989	0.499
tdc_pgp_broccatelli	classification	True	Uni-Mol V2 (84m)	0.933	TabPFNClassifier	0.927	0.625
chemml_cep_homo	regression	True	TabPFNRegressor	0.102	TabPFNRegressor	0.086	-19.231
chemml_organic_density	regression	True	TabPFNRegressor	0.005	TabPFNRegressor	0.005	5.907
esol_delaney	regression	True	MapLight CatBoost (Strict Parity)	0.645	TabPFNRegressor	0.612	-5.269
freesolv_sampl	regression	True	TabPFNRegressor	0.933	Chemprop v2 (AttentiveFP, ensemble=1)	1.080	13.562
lipophilicity	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.551	Uni-Mol V1	0.592	6.932
poduam_pod_nc_std	regression	True	Random forest	0.720	Extra trees	0.701	-2.720
poduam_pod_rd_std	regression	True	XGBoost (ADMETboost features)	0.567	XGBoost	0.551	-2.872
polaris_adme_fang_hppb_1	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.444	MapLight + GNN (CatBoost, Strict Parity)	0.449	1.072
polaris_adme_fang_perm_1	regression	True	Uni-Mol V1	0.399	Uni-Mol V1	0.399	-0.040
polaris_adme_fang_rclint_1	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.517	Uni-Mol V1	0.512	-0.968
polaris_adme_fang_rppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.493	MapLight + GNN (CatBoost, Strict Parity)	0.494	0.170
polaris_adme_fang_solu_1	regression	True	Uni-Mol V2 (84m)	0.551	Uni-Mol V1	0.575	4.202
tdc_caco2_wang	regression	True	MapLight CatBoost (Strict Parity)	0.350	MapLight CatBoost (Strict Parity)	0.350	0.000
tdc_clearance_hepatocyte_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	44.113	MapLight + GNN (CatBoost, Strict Parity)	43.976	-0.311
tdc_clearance_microsome_az	regression	True	Uni-Mol V2 (84m)	33.507	Uni-Mol V1	35.888	6.634

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Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_half_life_obach	regression	True	Uni-Mol V1	19.128	Uni-Mol V1	17.222	-11.067
tdc_ld50_zhu	regression	True	TabPFNRegressor	0.806	HistGradientBoosting	0.839	3.931
tdc_lipophilicity_astrazeneca	regression	True	Chemprop v2 (D-MPNN, ensemble=3)	0.577	Uni-Mol V1	0.587	1.698
tdc_ppbr_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	11.356	MapLight + GNN (CatBoost, Strict Parity)	11.444	0.762
tdc_solubility_aqsolddb	regression	True	TabPFNRegressor	0.961	Uni-Mol V1	0.989	2.807
tdc_vdss_lombardo	regression	True	MapLight + GNN (CatBoost, Strict Parity)	4.670	MapLight + GNN (CatBoost, Strict Parity)	4.668	-0.035
tdc_carcinogens_lagunin	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	AdaBoost	0.863	7.656
tdc_clintox	classification	False	Random forest	0.956	CatBoost	0.949	0.788
tdc_pampa_ncats	classification	False	Uni-Mol V1	0.763	Extra trees	0.805	-5.206
tdc_skin_reaction	classification	False	XGBoost (ADMETboost features)	0.667	TabPFNClassifier	0.747	-10.737
tdc_tox21	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	XGBoost	0.812	-26.451
tdc_toxcast	classification	False	HistGradientBoosting	0.716	CatBoost	0.791	-9.562
tdc_hydrationfreeenergy_freesolv	regression	False	TabPFNRegressor	1.116	TabPFNRegressor	0.592	-88.600

Table S6: Dataset catalog. “Ranking metric” is the metric used for cross-dataset model ranking (RMSE for regression; the dataset’s designated primary classification metric otherwise); leaderboard comparisons in Table S7 use the leaderboard’s own metric, which differs for several TDC regression tasks.

Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	QSARena			Best published	Best published model
											Leaderboard metric	(lb metric)	Est. rank		
tdc_ames	TDC	classification	7,278	5,821	1,457	predefined	raw	roc_auc	Ensemble (OOFS tacking (RidgeCV, 5-fold))	0.883	ROC_-AUC	0.883	2	0.912	QW-MTL
tdc_bbb_martins	TDC	classification	2,030	1,624	406	predefined	raw	roc_auc	Random forest	0.925	ROC_-AUC	0.925	2	0.941	MolGPS (3B)
tdc_bioavailability_ma	TDC	classification	640	512	128	predefined	raw	roc_auc	CatBoost	0.777	ROC_-AUC	0.777	1	0.748	MaxQsaring
tdc_carcinogens_lagunin	TDC	classification	280	223	57	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	ROC_-AUC	0.929	1	0.848	FATE-Tox (MTL)
tdc_clintox	TDC	classification	1,478	1,180	298	scaffold	raw	roc_auc	CFA (Combinatorial Fusion)	0.974	ROC_-AUC	0.974	3	0.996	PrismNet
tdc_cyp1a2_veith	TDC	classification	12,579	10,061	2,518	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse OOF error))	0.974					
tdc_cyp2c19_veith	TDC	classification	12,665	10,131	2,534	scaffold	raw	roc_auc	Ensemble (OOFS tacking (RidgeCV, 5-fold))	0.938					
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	669	534	135	predefined	raw	auprc	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	AUPRC	0.438	3	0.450	MaxQsaring
tdc_cyp2c9_veith	TDC	classification	12,092	9,673	2,419	predefined	raw	auprc	Ensemble (OOFS tacking (RidgeCV, 5-fold))	0.827	AUPRC	0.827	3	0.877	MaxQsaring
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	667	532	135	predefined	raw	auprc	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	AUPRC	0.693	4	0.766	MaxQsaring
tdc_cyp2d6_veith	TDC	classification	13,130	10,504	2,626	predefined	raw	auprc	Ensemble (OOFS tacking (RidgeCV, 5-fold))	0.763	AUPRC	0.763	2	0.811	MaxQsaring
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	670	535	135	predefined	raw	auprc	XGBoost (ADMETboost features)	0.729	ROC_-AUC	0.665	6	0.692	MolE

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSArena (lb metric)	Est. rank	Best published	Best published model
tdc_cyp3a4_veith	TDC	classification	12,328	9,861	2,467	predefined	raw	auprc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.905	AUPRC	0.905	2	0.923	MaxQsaring
tdc_dili	TDC	classification	475	379	96	predefined	raw	roc_auc	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.933	ROC_-AUC	0.933	4	0.945	Meta-model (NIST)
tdc_herg	TDC	classification	655	523	132	predefined	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	ROC_-AUC	0.857	3	0.880	MaxQsaring
tdc_herg_karim	TDC	classification	13,445	10,755	2,690	scaffold	raw	roc_auc	Ensemble (Weighted average (inverse OOF error))	0.907					
tdc_hia_hou	TDC	classification	578	461	117	predefined	raw	roc_auc	XGBoost (ADMETboost features)	0.994	ROC_-AUC	0.994	2	0.994	MiniMol (GINE)
tdc_pampa_ncats	TDC	classification	2,034	1,626	408	scaffold	raw	roc_auc	Uni-Mol V1	0.763					
tdc_pgp_broccatelli	TDC	classification	1,218	973	245	predefined	raw	roc_auc	Uni-Mol V2 (84m)	0.933	ROC_-AUC	0.933	4	0.994	MiniMol (GINE)
tdc_skin_reaction	TDC	classification	404	289	115	scaffold	raw	roc_auc	XGBoost (ADMETboost features)	0.667	ROC_-AUC	0.667	>10	0.741	FATE-Tox (MTL)
tdc_tox21	TDC	classification	7,258	5,797	1,461	scaffold	raw	roc_auc	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	ROC_-AUC	0.597	>10	0.867	PrismNet
tdc_toxcast	TDC	classification	1,731	1,357	374	scaffold	raw	roc_auc	HistGradient Boosting	0.716	ROC_-AUC	0.716	1	0.714	PrismNet
chemml_cep_homo	ChemML	regression	500	400	100	target_-	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.090					
chemml_organic_density	ChemML	regression	500	400	100	quartiles target_-	log10	rmse	TabPFNRegressor	0.005					
esol_delaney	MoleculeNet	regression	1,128	874	254	quartiles scaffold	raw	rmse	Ensemble (Weighted average (inverse OOF error))	0.622	RMSE	0.622	7	0.558	GCN
freesolv_sampler	MoleculeNet	regression	642	513	129	random	raw	rmse	TabPFNRegressor	0.933					
lipophilicity	MoleculeNet	regression	4,200	3,357	843	scaffold	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.507	RMSE	0.507	1	0.549	GCN

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	Leaderboard metric	QSArena (lb metric)	Est. rank	Best published	Best published model
poduam_pod_nc_std	PODUAM	regression	1,842	1,473	369	target_- raw	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.694	RMSE	0.694	2	0.550	PODUAM BNN (PODnc)
poduam_pod_rd_std	PODUAM	regression	2,355	1,884	471	quartiles target_- raw	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.556	RMSE	0.556	2	0.410	PODUAM BNN (PODr)
polaris_adme_fang_hppb_1	Polaris	regression	1,808	1,446	362	quartiles predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.422	MSE	0.178	4	0.143	LateFusion-k5
polaris_adme_fang_perm_1	Polaris	regression	2,642	2,113	529	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.379	MSE	0.144	4	0.113	1B_MPNN_MolGPS-ens_Large Mix
polaris_adme_fang_rclint_1	Polaris	regression	3,054	2,443	611	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.484	MSE	0.234	5	0.204	LateFusion-k5
polaris_adme_fang_rppb_1	Polaris	regression	885	708	177	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.478	MSE	0.228	1	0.230	1B_MPNN_LargeMix-and-Phenomics
polaris_adme_fang_solu_1	Polaris	regression	2,173	1,738	435	predefined	raw	rmse	Uni-Mol V2 (84m)	0.551	MSE	0.304	7	0.222	1B_MPNN_MolGPS-ens_Large Mix
tdc_caco2_wang	TDC	regression	910	728	182	predefined	raw	rmse	Ensemble (OOFS Stacking (RidgeCV, 5-fold))	0.338	MAE	0.269	3	0.256	CaliciBoost
tdc_clearance_hepatocyte_az	TDC	regression	1,213	970	243	predefined	raw	rmse	MapLight + GNN (CatBoost, Strict Parity)	44.113	SPEARMAN	0.516	3	0.633	CFA
tdc_clearance_microsome_az	TDC	regression	1,102	881	221	predefined	raw	rmse	Uni-Mol V2 (84m)	33.507	SPEARMAN	0.678	1	0.652	MapLight + GNN
tdc_half_life_obach	TDC	regression	667	532	135	predefined	raw	rmse	Uni-Mol V1	19.128	SPEARMAN	0.597	2	0.649	CFA
tdc_hydrationfreeenergy_freesolv	TDC	regression	642	490	152	scaffold	raw	rmse	Ensemble (Weighted average (inverse OOF error))	1.075	RMSE	1.075	7	0.654	PrismNet

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Dataset	Suite	Task	Molecules	Train	Test	Split	Target scale	Ranking metric	Best model	Best value	QSARena			Best published	Best published model
											Leaderboard metric	(lb metric)	Est. rank		
tdc_ld50_zhu	TDC	regression	7,385	5,907	1,478	predefined	draw	rmse	TabPFNRegressor	0.806	MAE	0.552	2	0.292	Base Boosting KyQVZ6b2
tdc_lipophilicity_astrazeneca	TDC	regression	4,200	3,360	840	predefined	draw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.551	MAE	0.412	3	0.406	MiniMol
tdc_ppbr_az	TDC	regression	2,790	2,231	559	predefined	draw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	10.768	MAE	7.118	2	0.679	Gradient Boost
tdc_solubility_aqsoldb	TDC	regression	9,980	7,985	1,995	predefined	draw	rmse	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.955	MAE	0.676	2	0.557	MiniMol
tdc_vdss_lombardo	TDC	regression	1,130	904	226	predefined	draw	rmse	MapLight + GNN (CatBoost, Strict Parity)	4.670	SPEARMAN	0.680	5	0.942	MapLight + GNN

Table S7: Per-dataset leaderboard comparison, sorted by estimated rank. “References (n)” is the number of published values available for that dataset and metric; ranks from sparse reference sets are correspondingly uncertain. The **tdc_ppbr_az** top-1 reference (MAE 0.679) is inconsistent in scale with the rest of that dataset’s references (top-10 cutoff 7.914) and its top-1 gap should be disregarded.

Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV-selected est. rank
lipophilicity	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.507	0.549	0.610	27	1	TabPFNRegressor	0.595	8
polaris_adme_fang_rppb_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.228	0.230	0.634	10	1	XGBoost (ADMETboost features)	0.257	3
tdc_bioavailability_ma	ROC_- AUC	CatBoost	0.777	0.748	0.640	8	1	XGBoost (ADMETboost features)	0.721	2
tdc_carcinogens_lagunin	ROC_- AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	0.848	0.795	20	1	XGBoost (ADMETboost features)	0.908	1
tdc_clearance_microsome_az	SPEARMA	AutoMol V2 (84m)	0.678	0.652	0.599	17	1	XGBoost (ADMETboost features)	0.549	>10
tdc_toxcast	ROC_- AUC	HistGradientBoosting	0.716	0.714	0.714	2	1	XGBoost (ADMETboost features)	0.704	3
poduam_pod_nc_std	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.694	0.550	0.730	2	2	XGBoost (ADMETboost features)	0.729	2
poduam_pod_rd_std	RMSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.556	0.410	0.630	2	2	XGBoost (ADMETboost features)	0.567	2
tdc_ames	ROC_- AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.883	0.912	0.834	7	2	Extra trees	0.871	2
tdc_bbb_martins	ROC_- AUC	Random forest	0.925	0.941	0.903	7	2	XGBoost (ADMETboost features)	0.916	2
tdc_cyp2d6_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.763	0.811	0.464	6	2	AdaBoost	0.615	6
tdc_cyp3a4_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.905	0.923	0.750	6	2	AdaBoost	0.798	6
tdc_half_life_obach	SPEARMA	AutoMol V1	0.597	0.649	0.485	16	2	XGBoost (ADMETboost features)	0.473	>10

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV- selected est. rank
tdc_hia_hou	ROC_AUC	XGBoost (ADMETboost features)	0.994	0.994	0.976	8	2	XGBoost (ADMETboost features)	0.994	2
tdc_ld50_zhu	MAE	TabPFNRegressor	0.552	0.292	0.605	16	2	XGBoost	0.588	8
tdc_ppbr_az	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	7.118	0.679	7.914	16	2	TabPFNRegressor	8.537	>10
tdc_solubility_aqsolddb	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.676	0.557	0.776	17	2	XGBoost (ADMETboost features)	0.719	2
tdc_caco2_wang	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.269	0.256	0.288	20	3	XGBoost (ADMETboost features)	0.293	>10
tdc_clearance_hepatocyte_az	SPEARMAN	MANLight + GNN (Cat Boost, Strict Parity)	0.516	0.633	0.440	16	3	XGBoost (ADMETboost features)	0.369	>10
tdc_clintox	ROC_AUC	CFA (Combinatorial Fusion)	0.974	0.996	0.889	12	3	AdaBoost	0.886	>10
tdc_cyp2c9_substrate_carbonmangels	AUPRC	Chemprop v2 (AttentiveFP, ensemble=3)	0.438	0.450	0.360	6	3	SVC	0.301	7
tdc_cyp2c9_veith	AUPRC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.827	0.877	0.770	6	3	AdaBoost	0.705	7
tdc_herg	ROC_AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.857	0.880	0.806	7	3	XGBoost (ADMETboost features)	0.792	8
tdc_lipophilicity_astrazeneca	MAE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.412	0.406	0.515	17	3	TabPFNRegressor	0.439	4
polaris_adme_fang_hppb_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.178	0.143	0.303	10	4	XGBoost (ADMETboost features)	0.214	5
polaris_adme_fang_perm_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.144	0.113	0.239	10	4	XGBoost (ADMETboost features)	0.194	8
tdc_cyp2d6_substrate_carbonmangels	AUPRC	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.693	0.766	0.570	7	4	ChemML MLP (PyTorch)	0.512	8
tdc_dili	ROC_AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.933	0.945	0.852	6	4	XGBoost (ADMETboost features)	0.923	4
tdc_pgp_broccatelli	ROC_AUC	Uni-Mol V2 (84m)	0.933	0.994	0.911	8	4	Random forest	0.904	9
polaris_adme_fang_rcint_1	MSE	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.234	0.204	0.337	10	5	XGBoost	0.293	6

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Dataset	Metric	QSARena best model (test-selected)	QSARena value	Reference top-1	Reference top-10 cutoff	References (n)	Est. rank	CV-selected model	CV-selected value	CV- selected est. rank
tdc_vdss_ lombardo	SPEARMAN	MAPLight + GNN (Cat Boost, Strict Parity)	0.680	0.942	0.582	16	5	XGBoost (ADMETboost features)	0.465	>10
tdc_cyp3a4_ substrate_ carbonmangels	ROC_ AUC	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.665	0.692	0.651	7	6	XGBoost (ADMETboost features)	0.660	7
esol_delaney	RMSE	Ensemble (Weighted average (inverse OOF error))	0.622	0.558	0.743	30	7	TabPFNRegressor	0.673	9
polaris_adme_ fang_solu_1	MSE	Uni-Mol V2 (84m)	0.304	0.222	0.323	10	7	XGBoost (ADMETboost features)	0.353	>10
tdc_ hydrationfreeenergy_ freesolv	RMSE	Ensemble (Weighted average (inverse OOF error))	1.075	0.654	1.211	12	7	XGBoost (ADMETboost features)	1.138	9
tdc_skin_reaction	ROC_ AUC	XGBoost (ADMETboost features)	0.667	0.741	0.677	21	>10	XGBoost (ADMETboost features)	0.667	>10
tdc_tox21	ROC_ AUC	Chemprop v2 (AttentiveFP, ensemble=3)	0.597	0.867	0.840	12	>10	Extra trees	0.549	>10

Table S8: Applicability domain and calibration on the 22 official TDC ADMET Benchmark Group splits, for a fixed reference model (random forest, 300 trees, on Morgan radius-2 2048-bit fingerprints plus RDKit 2D descriptors) fitted to 80% of each train_val partition, with the remaining 20% used only for conformal calibration. Coverage is the fraction of test compounds inside the domain. The error ratio is the mean test error outside the domain divided by that inside it (MAE for regression, 0/1 error for classification); values above 1 mean the domain flag separates worse predictions. “Std.” is the Roy standardization approach on RDKit descriptors; “kNN” is the 5-nearest-neighbour Tanimoto distance thresholded at the 95th percentile of training distances; “Reliability” is the conformal-interval-width flag (regression) or the posterior-probability confidence flag with $|p_1 - p_0| \geq 0.5$ (classification). Conformal coverage is the empirical test coverage of 90% split-conformal intervals or prediction sets. “—” marks a ratio not reported because fewer than five test compounds fall on one side of the domain boundary. Results describe the reference model, not the benchmark’s selected model; single seed.

Dataset	Task	n test	Std. coverage	Std. ratio	kNN coverage	kNN ratio	Reliability ratio	Conformal coverage	ECE
caco2_wang	reg.	182	0.99	—	0.74	0.75	1.38	0.95	—
clearance_hepatocyte_az	reg.	243	0.94	1.09	0.88	1.26	1.75	0.94	—
clearance_microsome_az	reg.	221	0.99	—	0.97	1.32	2.08	0.93	—
half_life_obach	reg.	135	0.95	1.36	0.94	0.87	3.60	0.94	—
ld50_zhu	reg.	1478	0.95	1.16	0.82	1.24	2.11	0.82	—
lipophilicity_astazeneca	reg.	840	0.99	2.33	0.94	1.55	1.47	0.88	—
ppbr_az	reg.	559	0.99	1.35	0.91	1.05	3.01	0.93	—
solubility_aqsolddb	reg.	1995	0.95	1.45	0.86	1.15	1.59	0.89	—
vdss_lombardo	reg.	226	0.94	0.49	0.86	1.22	8.26	0.95	—
ames	cls.	1457	0.96	0.91	0.84	1.71	2.67	0.89	0.07
bbb_martins	cls.	406	0.96	0.00	0.78	1.19	9.30	0.91	0.06
bioavailability_ma	cls.	128	0.98	—	0.96	0.72	3.72	0.91	0.08
cyp2c9_substrate_carbonmangels	cls.	135	0.93	0.68	0.95	0.00	2.15	0.81	0.11
cyp2c9_veith	cls.	2419	0.96	0.53	0.88	0.62	5.21	0.91	0.05
cyp2d6_substrate_carbonmangels	cls.	135	0.96	0.65	0.98	—	4.82	0.89	0.06
cyp2d6_veith	cls.	2626	0.96	0.90	0.84	0.89	5.29	0.91	0.06
cyp3a4_substrate_carbonmangels	cls.	135	0.95	2.08	0.94	1.80	1.47	0.93	0.06
cyp3a4_veith	cls.	2467	0.98	0.91	0.94	0.93	4.88	0.89	0.07
dili	cls.	96	0.95	3.64	0.95	0.00	2.16	0.95	0.17
herg	cls.	132	0.99	—	0.92	1.10	4.71	0.86	0.10
hia_hou	cls.	117	0.97	—	0.82	4.57	6.13	0.91	0.09
pgp_broccatelli	cls.	245	0.94	0.75	0.84	1.21	5.04	0.88	0.06

Table S9: Dataset meta-feature catalog computed on each benchmark train/test partition (verified against the recorded split hashes). Diversity and similarity use Morgan radius-2, 2,048-bit fingerprints; SNN is the maximum Tanimoto similarity of a test molecule to any training molecule, and OOD fraction is the share of test molecules with SNN below 0.40.

Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_ames	classification	patidefinedest_-	auc	5,821	1,457	0.533	1.142			2,387	0.410	0.809	0.916	0.517	0.251
			roc_-												
tdc_bbb_martins	classification	patidefinedest_-	auc	1,624	406	0.753	3.050			829	0.510	0.744	0.890	0.418	0.520
			roc_-												
tdc_bioavailability_ma	classification	patidefinedest_-	auc	512	128	0.771	3.376			363	0.709	0.837	0.899	0.435	0.492
			roc_-												
tdc_carcinogens_lagunin	classification	scaffolddest_-	auc	223	57	0.184	4.439			191	0.857	0.864	0.914	0.346	0.684
			roc_-												
tdc_clintox	classification	scaffolddest_-	auc	1,180	298	0.083	11.041			894	0.758	0.813	0.915	0.425	0.517
			roc_-												
tdc_cyp1a2_veith	classification	scaffolddest_-	auc	10,061	2,518	0.445	1.249			7,622	0.758	0.811	0.887	0.561	0.199
			roc_-												
tdc_cyp2c19_veith	classification	scaffolddest_-	auc	10,131	2,534	0.496	1.018			7,615	0.752	0.807	0.887	0.563	0.204
			roc_-												
			auc												

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_cyp2c9_-	classification	patidefined_-	auprc	534	135	0.193	4.184			366	0.685	0.831	0.899	0.408	0.504
substrate_-															
carbonmangels															
tdc_cyp2c9_-	classification	patidefined_-	auprc	9,673	2,419	0.339	1.954			5,727	0.592	0.784	0.885	0.509	0.338
veith															
tdc_cyp2d6_-	classification	patidefined_-	auprc	532	135	0.278	2.595			373	0.701	0.845	0.898	0.426	0.526
substrate_-															
carbonmangels															
tdc_cyp2d6_-	classification	patidefined_-	auprc	10,504	2,626	0.197	4.072			6,643	0.632	0.779	0.884	0.487	0.364
veith															
tdc_cyp3a4_-	classification	patidefined_-	auprc	535	135	0.516	1.066			375	0.701	0.832	0.899	0.436	0.474
substrate_-															
carbonmangels															
tdc_cyp3a4_-	classification	patidefined_-	auprc	9,861	2,467	0.408	1.448			6,088	0.617	0.800	0.889	0.534	0.260
veith															
tdc_dili	classification	patidefined_-	auc	379	96	0.491	1.038			265	0.699	0.879	0.907	0.332	0.781
			roc_-												
tdc_herg	classification	patidefined_-	auc	523	132	0.677	2.095			341	0.652	0.792	0.895	0.484	0.379
			roc_-												
tdc_herg_karim	classification	patidefined_-	auc	10,755	2,690	0.492	1.031			5,884	0.547	0.646	0.878	0.672	0.084
			roc_-												
			auc												

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_hia_hou	classification	patidefinedest_-	auc roc_-	461	117	0.889	8.039			317	0.688	0.868	0.894	0.371	0.598
tdc_pampa_ncats	classification	scaffolddest_-	auc roc_-	1,626	408	0.846	5.504			1,355	0.833	0.862	0.871	0.642	0.064
tdc_pgp_broccatelli	classification	patidefinedest_-	auc roc_-	973	245	0.541	1.177			546	0.561	0.764	0.874	0.490	0.400
tdc_skin_reaction	classification	scaffolddest_-	auc roc_-	289	115	0.661	1.949			247	0.855	0.895	0.918	0.312	0.904
tdc_tox21	classification	scaffolddest_-	auc roc_-	5,797	1,461	0.049	19.412			3,853	0.665	0.872	0.925	0.467	0.305
tdc_toxcast	classification	scaffolddest_-	auc roc_-	1,357	374	0.262	2.812			1,002	0.738	0.889	0.926	0.439	0.422
chemml_cep_homo	regression	target_test_-	auc rmse	400	100			0.687	0.029	400	1	1	0.844	0.492	0.220
chemml_organic_density	regression	target_test_-	auc rmse	400	100			0.031	-0.093	399	0.998	0.997	0.841	0.463	0.250
esol_delaney	regression	scaffolddest_-	auc rmse	874	254			2.247	-0.435	583	0.667	0.873	0.930	0.437	0.417
freesolv_sampler	regression	randomtest_-	rmse	513	129			3.677	-0.753	308	0.600	0.932	0.923	0.570	0.163

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
lipophilicity	regression	scaffold	test_rmse	3,357	843			1.190	-0.557	2,355	0.702	0.791	0.882	0.584	0.152
poduam_nc_std	regression	scaffold	test_rmse	1,473	369			1.109	-1.529	945	0.642	0.899	0.922	0.521	0.322
poduam_nc_std	regression	quartiles	test_rmse	1,884	471			0.865	-1.743	1,237	0.657	0.905	0.917	0.549	0.268
polaris_adme_fang_hppb_1	regression	undefined	test_rmse	1,446	362			0.719	-0.337	1,003	0.694	0.827	0.878	0.583	0.331
polaris_adme_fang_perm_1	regression	undefined	test_rmse	2,113	529			0.682	0.746	1,885	0.892	0.926	0.875	0.443	0.465
polaris_adme_fang_rclint_1	regression	undefined	test_rmse	2,443	611			0.751	-0.134	2,146	0.878	0.922	0.876	0.444	0.435
polaris_adme_fang_rppb_1	regression	undefined	test_rmse	708	177			0.751	-0.435	569	0.804	0.872	0.877	0.511	0.429
polaris_adme_fang_solu_1	regression	undefined	test_rmse	1,738	435			0.668	-1.657	1,531	0.881	0.925	0.874	0.446	0.492
tdc_caco2_wang	regression	undefined	test_rmse	728	182			0.797	-0.715	402	0.552	0.694	0.890	0.459	0.473
tdc_clearance	regression	undefined	test_rmse	970	243			50.287	1.216	601	0.620	0.687	0.877	0.425	0.580
hepatocyte_az	regression	undefined	test_rmse	881	221			45.128	1.548	587	0.666	0.818	0.871	0.520	0.321
microsome_az															

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Dataset	Task	Split	Metric	n train	n test	Positive fraction	Imbalance ratio	Target SD	Target skew	Scaffolds	Scaffolds / molecule	Singleton scaffold fraction	Internal diversity	Mean SNN	OOD fraction (SNN < 0.40)
tdc_half_life_obach	regression	undefined	rmse	532	135			90.920	9.338	355	0.667	0.820	0.899	0.413	0.533
tdc_hydrationfreeenergy_freesolv	regression	scaffold	rmse	490	152			4.014	-1.427	381	0.778	0.934	0.927	0.384	0.625
tdc_ld50_zhu	regression	undefined	rmse	5,907	1,478			0.921	0.928	3,136	0.531	0.894	0.917	0.448	0.438
tdc_lipophilicity	regression	undefined	rmse	3,360	840			1.204	-0.582	1,932	0.575	0.769	0.879	0.575	0.219
astrazeneca_tdc_ppbr_az	regression	undefined	rmse	2,231	559			18.785	-1.962	946	0.424	0.498	0.879	0.520	0.349
tdc_solubility_aqsolddb	regression	undefined	rmse	7,985	1,995			2.369	-0.587	4,265	0.534	0.900	0.922	0.475	0.327
tdc_vdss_lombardo	regression	undefined	rmse	904	226			24.829	24.663	613	0.678	0.822	0.900	0.384	0.619

Table S10: Exploratory family $\times$ meta-feature associations: Spearman correlation between each meta-feature and the relative gap of the family’s best model to the per-dataset best, with a dataset-bootstrap 95% CI (10,000 resamples), a two-sided label-permutation p-value and a Benjamini–Hochberg q-value over the whole grid. Positive ρ means the family falls further behind as the meta-feature increases.

Model family	Meta-feature	Datasets	Spearman rho	95% CI	Permutation p	BH q
Conventional ML	log10_n_train	44	0.034	-0.29 to 0.35	0.818	0.873
Conventional ML	internal_diversity	44	-0.329	-0.59 to -0.01	0.028	0.238
Conventional ML	mean_snn	44	0.325	0.01 to 0.58	0.034	0.238
Conventional ML	ood_fraction	44	-0.326	-0.59 to -0.01	0.035	0.238
Conventional ML	scaffolds_per_molecule	44	0.139	-0.20 to 0.43	0.367	0.710
Conventional ML	label_asymmetry	44	0.204	-0.10 to 0.46	0.179	0.461
Ensemble (stacking / averaging)	log10_n_train	44	-0.467	-0.70 to -0.19	0.001	0.058
Ensemble (stacking / averaging)	internal_diversity	44	0.207	-0.14 to 0.52	0.180	0.461
Ensemble (stacking / averaging)	mean_snn	44	-0.393	-0.66 to -0.09	0.008	0.202
Ensemble (stacking / averaging)	ood_fraction	44	0.288	-0.03 to 0.59	0.062	0.297
Ensemble (stacking / averaging)	scaffolds_per_molecule	44	0.147	-0.14 to 0.43	0.336	0.710
Ensemble (stacking / averaging)	label_asymmetry	44	0.275	-0.03 to 0.54	0.070	0.308
CFA combinatorial fusion	log10_n_train	44	-0.220	-0.47 to 0.07	0.156	0.461
CFA combinatorial fusion	internal_diversity	44	-0.325	-0.59 to -0.01	0.033	0.238
CFA combinatorial fusion	mean_snn	44	0.125	-0.21 to 0.43	0.421	0.710
CFA combinatorial fusion	ood_fraction	44	-0.081	-0.39 to 0.24	0.602	0.743
CFA combinatorial fusion	scaffolds_per_molecule	44	0.216	-0.10 to 0.50	0.159	0.461
CFA combinatorial fusion	label_asymmetry	44	0.108	-0.19 to 0.40	0.486	0.710
Uni-Mol (3D pretrained)	log10_n_train	44	-0.108	-0.41 to 0.20	0.488	0.710
Uni-Mol (3D pretrained)	internal_diversity	44	0.146	-0.21 to 0.49	0.348	0.710
Uni-Mol (3D pretrained)	mean_snn	44	0.081	-0.25 to 0.40	0.604	0.743
Uni-Mol (3D pretrained)	ood_fraction	44	-0.118	-0.46 to 0.22	0.442	0.710
Uni-Mol (3D pretrained)	scaffolds_per_molecule	44	0.102	-0.23 to 0.41	0.516	0.729
Uni-Mol (3D pretrained)	label_asymmetry	44	0.075	-0.26 to 0.41	0.628	0.753
TabPFN (tabular foundation)	log10_n_train	41	0.149	-0.24 to 0.50	0.346	0.710
TabPFN (tabular foundation)	internal_diversity	41	-0.242	-0.54 to 0.10	0.130	0.461
TabPFN (tabular foundation)	mean_snn	41	0.098	-0.21 to 0.40	0.540	0.730
TabPFN (tabular foundation)	ood_fraction	41	-0.004	-0.34 to 0.33	0.983	0.983
TabPFN (tabular foundation)	scaffolds_per_molecule	41	0.071	-0.28 to 0.41	0.661	0.755
TabPFN (tabular foundation)	label_asymmetry	41	-0.126	-0.42 to 0.18	0.428	0.710
Chemprop v2 GNN	log10_n_train	42	-0.008	-0.31 to 0.30	0.965	0.983
Chemprop v2 GNN	internal_diversity	42	-0.050	-0.37 to 0.28	0.752	0.840
Chemprop v2 GNN	mean_snn	42	0.131	-0.19 to 0.43	0.405	0.710
Chemprop v2 GNN	ood_fraction	42	-0.119	-0.44 to 0.21	0.454	0.710

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Model family	Meta-feature	Datasets	Spearman rho	95% CI	Permutation p	BH q
Chemprop v2 GNN	scaffolds_per_molecule	42	0.141	-0.16 to 0.42	0.370	0.710
Chemprop v2 GNN	label_asymmetry	42	0.325	-0.03 to 0.62	0.031	0.238
MapLight + GNN	log10_n_train	44	-0.205	-0.47 to 0.10	0.182	0.461
MapLight + GNN	internal_diversity	44	0.091	-0.23 to 0.40	0.559	0.730
MapLight + GNN	mean_snn	44	-0.071	-0.38 to 0.24	0.643	0.753
MapLight + GNN	ood_fraction	44	-0.113	-0.42 to 0.22	0.467	0.710
MapLight + GNN	scaffolds_per_molecule	44	0.291	0.01 to 0.54	0.052	0.278
MapLight + GNN	label_asymmetry	44	-0.090	-0.39 to 0.25	0.563	0.730
Deep tabular NN (ChemML MLP)	log10_n_train	44	-0.315	-0.57 to 0.00	0.042	0.250
Deep tabular NN (ChemML MLP)	internal_diversity	44	-0.251	-0.51 to 0.04	0.100	0.401
Deep tabular NN (ChemML MLP)	mean_snn	44	-0.017	-0.33 to 0.31	0.912	0.952
Deep tabular NN (ChemML MLP)	ood_fraction	44	0.039	-0.31 to 0.36	0.807	0.873
Deep tabular NN (ChemML MLP)	scaffolds_per_molecule	44	0.224	-0.10 to 0.51	0.141	0.461
Deep tabular NN (ChemML MLP)	label_asymmetry	44	0.181	-0.14 to 0.47	0.242	0.581

Table S11: Family selector (v2), leave-one-dataset-out regret: the relative gap (%) of the recommended family to the per-dataset best. Rows are pre-specified variants (model | feature blocks: F0 size, similarity, label asymmetry and task; F1 chemistry; F2 label landscape; F3 training-set CV landmarks); “sbs” always picks the family with the best record on the other datasets. The nested row chooses the variant inside an inner leave-one-dataset-out loop and is the only unbiased estimate; single variants are exploratory.

Selector (model feature blocks)	Mean regret (%)	Median regret (%)	Pick within 5% of best (%)	SBS-to-oracle gap closed
sbs	3.310	0.031	84.091	0
knn F0	3.760	0.159	79.545	-0.136
ridge F0	2.982	0	86.364	0.099
rf F0	4.236	0.082	77.273	-0.280
knn F0+F1	3.629	0	81.818	-0.096
ridge F0+F1	3.403	0	81.818	-0.028
rf F0+F1	3.650	1.086	81.818	-0.102
knn F0+F2	3.728	0.159	81.818	-0.126
ridge F0+F2	3.323	0	84.091	-0.004
rf F0+F2	4.163	0.159	75	-0.257
knn F0+F3	2.739	0	88.636	0.173
ridge F0+F3	3.133	0	84.091	0.054
rf F0+F3	3.216	0	84.091	0.028
knn F0-F3	3.375	0	81.818	-0.020
ridge F0-F3	4.525	0.031	81.818	-0.367
rf F0-F3	3.516	0.051	81.818	-0.062
tree_cls F0	4.837	0	79.545	-0.461
nested (headline)	3.115	0	86.364	0.059